

Stella's Architecture: Can a Small Decoder Replace a 7B+ LLM in an Encoder→World-Model→Decoder Stack?

TL;DR

- **Yes, the architecture exists and has been demonstrated — but never (publicly) at full conversational scale.** The two most directly relevant precedents are *Somniloquy* / *"Translating Latent State World Model Plans into Natural Language"* (Barker & Leonetti, RLDM 2025), which bolts a tiny ~few-million-parameter Transformer encoder-decoder head onto a DreamerV3 RSSM in Crafter, and *"The Mouth is Not the Brain"* (Niimi, arXiv:2601.17094, Jan 2026), which pairs a ~31K-parameter Deep Boltzmann Machine "world model" with a *frozen* GPT-2 "mouth" via a soft-prompt adapter and beats a fully fine-tuned GPT-2 baseline on review-generation cross-entropy (3.32 vs 4.74).
- **No one has published a true end-to-end conversational JEPA where a ~200M-param RSSM talks via a ~125M decoder without any larger LM in the loop.** The closest cousin is Meta's *Large Concept Model* (LCM) — but LCM still uses a SONAR encoder/decoder pair initialised from NLLB-1B (~1B params), and its "reasoning" model is 1.6B-7B, not an RSSM. The Stella spec is essentially an untrodden region of the design space, which is both an opportunity and a risk.
- **Recommended path: build on the Somniloquy pattern (RSSM + small Transformer decoder, language gradients detached), but warm-start the decoder from a TinyStories-style or BART-base (~140M) checkpoint, and budget for the very real possibility that ≤125M decoders collapse on open-domain dialogue.** Eldan & Li's TinyStories paper (arXiv:2305.07759) showed coherent English is possible "below 10 million total parameters" only on a synthetic vocabulary restricted to what 3-to-4-year-olds use; on open-domain text, vanilla 125M GPT-2 is documented to "rarely generate coherent and consistent English text beyond a few words." A SONAR decoder or a BART-base (~140M) is a more defensible "mouth" than training a 125M decoder from scratch. (Hugging Face + 2)

Key Findings

1. Direct precedents for "WM speaks via small decoder, no LLM intermediary"

Somniloquy (Barker & Leonetti, RLDM 2025) is the closest existing implementation to Stella's intended architecture. It plugs a small Transformer encoder-decoder ("translator head") on top of a DreamerV3 RSSM. From the public `configs.yaml`, the translator head is `attention_heads: 2, encoder_blocks: 2, decoder_blocks: 2, token_embed_size: 256, dec_max_length: 150, use_bottleneck: True` — i.e., a few-million-parameter head, *much* smaller than even GPT-2-small. Crucially, the RLDM configuration sets language gradients to NOT flow back into the RSSM (README: *"by default, the translation gradients do not flow into the RSSM to update the learned latent representation"*), so the RSSM is shaped purely by the standard Dreamer losses (reconstruction, reward, continuation, KL) and the decoder learns a separate map $z_t \rightarrow \text{English}$. The verbalisation is exactly the "mouth" pattern Stella wants: *"the*

agent uses the latent world model to plan a sequence of latent states and actions ... the latent state sequence is passed to a translation component that translates these latent states into a natural language description ... what these states mean in terms of the agent's expected interaction with the environment." The metric they report (`mean_imagined_bleu_score` on Crafter) confirms the head is trained to high BLEU on imagined rollouts, but the language is task-narrated description, not free conversation. (`GitHub + 3`)

"The Mouth is Not the Brain" (Niimi, Jan 2026) is the most explicit architectural validation of Stella's philosophy — and it explicitly cites LeCun's critique of autoregressive LLMs as the motivation. The architecture is: a Deep Boltzmann Machine (DBM) with `v ∈ {0,1}^160`, L=2 hidden layers, ~31,000 total parameters → an MLP adapter producing K=16 soft-prompt embeddings → a **fully frozen GPT-2**. On 55,000 Amazon smartphone reviews, results are unambiguous: (`arxiv`)

Model	CE Loss	Cosine sim (SBERT all-MiniLM-L6-v2)
DBM → Adapter → frozen GPT-2 (proposed)	3.32	0.43
Direct MLP → frozen GPT-2 (no world model)	3.52	0.39
Frozen in-context learning (no training)	3.59	0.38
Full GPT-2 fine-tune	4.74	0.40

The paper's verbatim conclusion is critical for Stella: *"simple prompts lack expressiveness while detailed prompts cause output collapse in small LLMs"* — i.e., the soft-prompt adapter from the world model is what prevents the small frozen LM from collapsing. And: *"the fact that sufficient linguistic quality was achieved with GPT-2 suggests that, for structured domains, world understanding may be a greater bottleneck than language generation capacity."* This is, almost verbatim, Stella's design thesis. (`arXiv + 2`)

2. Adjacent precedents (informative but not direct)

- **Dynalang** (Lin et al., ICML 2024 oral) is the canonical RSSM-meets-language work. It extends DreamerV3 to a *multimodal* (text + image) world model — text tokens are part of the observation, not the agent's output channel. Dynalang demonstrates that an RSSM can be pretrained on text-only data without actions/rewards and that *token-level* (not sentence-level) representations substantially outperform alternatives. The LangRoom environment includes a proof-of-concept where Dynalang generates short language outputs grounded in environment state. Stella can adopt the Dynalang RSSM directly; the open problem is the *output* head, which Dynalang does only at the single-token grid-world scale, not at conversational complexity. (`OpenReview + 2`)
- **Meta's Large Concept Model (LCM, Dec 2024) + SONAR** (Duquenne et al., `arXiv:2308.11466`) is the most production-ready "decoder-as-mouth" pipeline. SONAR is a 1024-d fixed-size sentence embedding space; the SONAR paper states it used *"NLLB 1B*

dense model as initialization" for the full encoder-decoder pair, and the community-converted text decoder card on Hugging Face (raxtemur/SONAR_200_text_decoder) lists *"Model size: 1B params"*. LCM is a sequence-to-sequence model *in concept (sentence) space*. The Meta technical report explicitly notes: *"To some extent, the LCM architecture resembles the Jepa approach (LeCun, 2022) that also aims to predict the representation of the next observation in an embedding space."* For Stella, SONAR provides a battle-tested encoder and decoder — but at ~1B params per side, not 125M. There is no published 125M-class SONAR-quality decoder. (arxiv + 2)

- **vec2text (Morris et al., EMNLP 2023)** demonstrates a T5-base (~220M) decoder + iterative corrector can reconstruct text from sentence embeddings with 92% exact-match on 32-token Wikipedia inputs (BLEU 97.3). According to a literature-review summary, vec2text was trained on the MS MARCO dataset with embeddings totaling *"~350M tokens total (32 or 128 tokens)"* — i.e., on the order of millions of (text, embedding) pairs, not hundreds of millions. This is the strongest evidence that small decoders *can* map a single fixed-size vector to fluent text — but for *reconstruction*, not for free-form conversation. (Liner)
- **VLWM (Meta FAIR, Sept 2025)** — Chen et al.'s Vision-Language World Model writes plans as plain English text rather than opaque vectors, with a 1B critic outperforming 8B VLM baselines on WorldPrediction. This validates "plans expressed in language" but uses a 7B-class LM, not a tiny decoder. (Apolo) (Apolo)
- **LED-WM (Nguyen et al., Nov 2025)** drops DreamerV3's pixel decoder in favour of an entirely language-aware latent dynamics model for MESSENGER — but it does *not* generate language output, only consumes it. (Emergent Mind)
- **Language World Models (Cowen-Rivers & Naradowsky, 2020)** — early demonstration that language messages can be interpreted by predicting future latent observations rather than tokens. Conceptual parent of Dynalang. (arxiv)
- **SAGE (Mar 2025)** uses discrete latent variables as a state-action space for dialogue, enabling state-level rather than token-level RL — close in spirit but the underlying generator is still an LLM.

3. Is "text JEPA / conversational JEPA" a real thing?

Mostly no, in the strict sense. The JEPA family is overwhelmingly vision-oriented (I-JEPA, V-JEPA, V-JEPA 2, LeJEPA, LeWorldModel, MC-JEPA, Audio-JEPA, Point-JEPA). The text variants that exist are:

- **Text-JEPA (Springer, 2025)** — applies JEPA self-supervised pretraining to *NL-to-First-Order-Logic conversion*, not dialogue. (Springer)
- **TI-JEPA (Mar 2025)** — text-image multimodal energy-based JEPA, alignment-focused, no generation. (arXiv)
- **TC-JEPA (Apple, Oct 2025)** — text-conditioned JEPA for *visual* representation learning. (Apple Machine Learning R...)
- **JEPA-T (Oct 2025)** — text-fused JEPA for *image* generation. (arxiv) (Hugging Face)
- **T-JEPA** — tabular data JEPA, unrelated to text generation. (OpenReview)

There is no published "conversational JEPA" or "dialogue-JEPA" paper. The LCM is the closest in spirit, and Meta explicitly frames it as JEPA-adjacent.

4. How small can a coherent open-domain language decoder be?

The honest answer from the published literature is **not as small as 125M for open-domain conversation, unless the domain is constrained or the model is conditioned by a world model**:

- **TinyStories** (Eldan & Li, arXiv:2305.07759) showed Transformers "below 10 million total parameters" can produce *"fluent and consistent stories with several paragraphs that are diverse and have almost perfect grammar"* — but only on a synthetic vocabulary restricted to what a 3-to-4-year-old typically understands. Their paper's framing sentence is exactly the warning Stella needs: *"Models with around 125M parameters such as GPT-Neo (small) or GPT-2 (small) can rarely generate coherent and consistent English text beyond a few words even after extensive training."* The 33M figure widely cited online is just the largest model in their released checkpoint series, not the paper's threshold claim. (Hugging Face + 3)
- **"Tiny Transformers Excel at Sentence Compression"** (Belcak & Wattenhofer, 2024) shows 1-3-layer transformers can encode/decode standard English sentences from a single ~3kB token — strong evidence the *information-theoretic* room exists. (arxiv)
- **Sentence Bottleneck Autoencoders** (Montero et al., EMNLP 2021) — single-layer decoder on top of a frozen RoBERTa-base produces coherent sentence-level outputs.
- **Niimi 2026 (Mouth Is Not the Brain)** is the most direct counterexample: a *frozen* GPT-2 becomes coherent on Amazon reviews *because* it is conditioned by world-model soft prompts; without that conditioning, *"detailed prompts cause output collapse."* (arxiv)

The implication for Stella: **a 125M decoder trained from scratch on open-domain dialogue is the riskiest possible choice**. The two robust options are (a) freeze a small pretrained LM (GPT-2-124M, OPT-125M, Pythia-160M, or BART-base-140M) and condition it via adapter / soft prompt from the world model — exactly the Mouth-is-not-the-Brain recipe — or (b) accept a ~1B SONAR-class decoder.

5. Ablations comparing WM+small-decoder vs WM+large-LLM

Direct head-to-head ablations of the type Stella wants barely exist in the literature. The Niimi 2026 ablation table above is the only published direct comparison of (world-model + tiny adapter + small frozen LM) vs (no-world-model baselines) on the same data, and it favors the world-model conditioning — but it does NOT include a 7B+ LLM arm. The LCM paper (Meta, Dec 2024) reports that *in zero-shot tests on the XLSum benchmark, a 7B LCM operating in SONAR space outperformed Llama-3.1-8B* (ROUGE-L across languages, per the LCM technical report) — suggestive evidence that the "concept-space" pipeline can match a token-space LLM at equal parameter count, but both arms are 7B-class and the comparison is zero-shot only. **The 125M-vs-7B ablation Stella is implicitly betting on has not been published**. This is itself a publishable contribution. (InfoQ)

Details

Open-source implementations Stella can actually fork

Repo	What it gives	Stella relevance
<code>m-barker/somniloquy-dreamer-v3</code> (MIT, RLDM 2025)	PyTorch DreamerV3 + translator head, configs for Crafter and MiniGrid, <code>github</code> BLEU eval scaffolding	Highest direct relevance. Architecture matches Stella's spec. Translator head is far smaller than 125M; you can scale it up.
<code>jlin816/dynalang</code> (ICML 2024)	JAX DreamerV3-derived multimodal RSSM, <code>GitHub</code> token-level text obs/output	Use as the RSSM/encoder backbone; replace its image branch with sentence-transformer embeddings.
<code>facebookresearch/SONAR</code> (MIT, model weights mixed MIT/NC)	200-language sentence encoder + decoder, fixed 1024-d embedding space	Drop-in encoder for <code>z_t</code> input; drop-in decoder if you accept ~1B params for the "mouth."
<code>facebookresearch/large_concept_model</code> (MIT)	Reference LCM training code in SONAR space (MSE, diffusion, two-tower variants)	Reference for sentence-space sequence modelling; useful for hybrid "RSSM-in-concept-space."
<code>vec2text/vec2text</code> (MIT)	T5-base hypothesizer + corrector for embedding inversion	Pretrained text-from-embedding decoder you can warm-start from.

What the Niimi 2026 architecture looks like, mapped onto Stella

Niimi component	Stella analog	Param count Niimi used	Stella target
DBM world model	RSSM	~31K params	~200M params
Adapter MLP → K=16 soft prompts	RSSM-state-to-decoder projection	Small MLP	Small MLP / cross-attention
Frozen GPT-2	Frozen small LM ("mouth")	GPT-2 (frozen, all params)	~125M target (frozen or adapter-tuned)

Niimi component	Stella analog	Param count Niimi used	Stella target
Visible input: 160-d categorical	sentence-transformer z_t (e.g., 384-1024-d)	160-d	384-768-d

The fit is almost one-to-one. The single most important architectural insight is Niimi's: **don't fine-tune the small LM end-to-end** ("B2: Fine-tuned LLM" was the worst baseline at CE 4.74 — "severe overfitting"). Train the world model and the adapter; keep the decoder frozen or only LoRA-tuned. [\(arxiv\)](#)

VRAM budget reality check on RX 7900 XTX (24GB)

- Llama-3-17GB (current Stella): ~17GB inference, leaves ~7GB headroom — tight.
- Proposed: 200M RSSM (~0.8GB fp16) + 125M decoder (~0.5GB fp16) + 384-d sentence-transformer encoder (~0.5GB) + KV/activation overhead → comfortably <4GB inference. Goal of <8GB is easily met; you have budget to go to a 350M decoder (BART-large) or even a ~1B SONAR decoder and still beat 8GB.

What the world model actually adds in conversation

Stella's thesis — that the WM is the "brain" — is supported by exactly one published controlled experiment (Niimi 2026), and only in product-review generation. The mechanism is concrete: the WM's energy function distinguishes coherent from incoherent state configurations (Niimi: *"the DBM's energy function distinguishes coherent from incoherent market configurations, assigning higher energy to implausible brand-price combinations"*), and the soft-prompt adapter routes this structured belief into the LM. For dialogue, the analog is: the RSSM should maintain belief over conversational state (user intent, sentiment trajectory, factual claims made so far) and route this into the decoder. *This is exactly what current LLMs do badly* (Niimi cites hallucination as the symptom). Whether an RSSM can learn this representation from conversational data alone is the open empirical question. [\(arxiv\)](#)

Recommendations

Stage 1 — Replicate Somniloquy on conversation (2-4 weeks). Fork [\(m-barker/somniloquy-dreamer-v3\)](#). Replace the Crafter observation encoder with a sentence-transformer (e.g., [\(all-MiniLM-L6-v2\)](#), 22M params, 384-d). Replace the translator head with a slightly larger Transformer decoder (still ≤50M params) trained to map RSSM latent states to *response text* on a small open-domain dialogue set (DailyDialog, PersonaChat). Keep [\(language_grads=False\)](#) so the RSSM is shaped only by reward + reconstruction. **Success benchmark:** the decoder produces grammatical 1-2 sentence responses whose semantic content tracks dialogue state (measure with SBERT cosine to ground-truth response ≥ 0.4, matching Niimi's threshold).

Stage 2 — Apply the Mouth-is-not-the-Brain pattern (4-8 weeks). If Stage 1 plateaus on fluency, swap the from-scratch decoder for a frozen pretrained 125M LM (GPT-2-small, Pythia-160M, or BART-base which has a real encoder-decoder structure and was designed for

conditional generation). Train an MLP adapter from RSSM state $\rightarrow$ K soft-prompt tokens (Niimi used K=16). Do **not** fine-tune the LM. **Success benchmark:** CE loss improves vs unconditioned LM by ≥ 0.2 nats (Niimi observed 0.27), and human-judged coherence $\geq 80\%$.

Stage 3 — Publishable ablation (the contribution that doesn't exist yet). Compare four arms on the same dialogue benchmark: (a) Llama-3-17GB baseline; (b) frozen 125M LM alone; (c) frozen 125M LM + RSSM-soft-prompt (your Stage 2); (d) frozen ~ 1 B SONAR decoder + RSSM. **This is the 125M-vs-7B ablation no one has published.** If (c) lands within 80% of (a)'s coherence at $< 5\%$ of (a)'s VRAM, that is a publishable result and validates Stella's bet.

Thresholds that should change the plan:

- If Stage 1 produces ungrammatical output even after 100k steps $\rightarrow$ **abandon from-scratch decoder**, jump to Stage 2 immediately.
- If Stage 2 still hallucinates as badly as the baseline LM $\rightarrow$ the RSSM is not capturing dialogue state. Switch RSSM input from sentence-transformer to *token-level* embeddings (Dynalang's finding: "*token-level representations substantially outperform*" sentence-level). (arXiv)
- If Stage 3 (c) lags Llama by $> 25\%$ on coherence even with a ~ 1 B SONAR decoder $\rightarrow$ the "125M decoder" target is too aggressive; settle on 350M-1B as the floor.

Stack to start from today:

- RSSM: DreamerV3 PyTorch implementation (Somniloquy's `m-barker/somniloquy-dreamer-v3`)
- Encoder: `sentence-transformers/all-MiniLM-L6-v2` or `BAAI/bge-small-en`
- Decoder option A (from-scratch, ~ 30 M): TinyStories-33M architecture
- Decoder option B (frozen LM, ~ 140 M): `facebook/bart-base` — purpose-built for encoder $\rightarrow$ decoder conditional generation and runs comfortably with adapters
- Decoder option C (SONAR, ~ 1 B): `facebook/SONAR` text decoder

Caveats

1. **No one has shipped this for open-domain dialogue.** Every directly-relevant precedent (Somniloquy, Mouth-is-not-the-Brain, Dynalang, LWM, LED-WM) operates on a constrained domain — Crafter, MiniGrid, MESSENGER, Amazon reviews, embodied QA. Stella is extrapolating outside the demonstrated envelope.
2. **Niimi 2026 is an arXiv preprint from January 2026 and has not yet been peer reviewed at the time of writing.** Its specific numerical claims (CE 3.32, etc.) should be treated as preliminary. The architectural insight (separate world model from frozen LM via soft prompt) is the durable part.
3. **The Somniloquy decoder is *not* trained as a conversational agent.** It is trained to narrate planned trajectories — short, templated language. Scaling to free-form dialogue is a research jump, not an engineering exercise.

4. **The widely cited claim that 125M parameters cannot produce coherent open-domain English comes from Eldan & Li 2023** (TinyStories) and is essentially a negative result about *unconditional* small LMs. With strong conditioning from a world model (as in Niimi), the operating regime may be different — but this has only been shown in one domain.
5. **"Conversational JEPA" as a published architecture does not exist.** Stella would be coining the term. The closest published kin is LCM, which Meta itself describes as JEPA-adjacent.
6. **The LeJEPA / LeWorldModel line (2025-2026) is rapidly evolving** and may obsolete some of the above before Stella ships; monitor Balestriero & LeCun's group output.